Experiments

All runs are reproducible from the commands in `run_pipeline.sh`. Their parameters are recorded in `configs/*.yaml` and in each run's `experiments/<M>/<run>/run_config.json`. Metrics are in `experiments/benchmark_eval_*` and `outputs/statistics/`. See `docs/results.md` for the numbers.

Hardware and software

Apple M4 (10-core CPU, 10-core GPU), 16 GB unified memory, macOS 26.2. Python 3.11, TensorFlow 2.15.1, tensorflow-metal 1.1.0. Full lock file: `requirements.lock.txt`. GlaViTU commit `8c4eead670ef64764f8db2708bba3c8bbbfc67bb`.

- **Inference:** CPU. tensorflow-metal intermittently raises SIGBUS inside `model.predict`.
- **Training:** Metal GPU, 4.5× faster than CPU (2.4 s vs 10.8 s per step at batch 4, identical losses to 6 decimals). Training is wrapped in `train_with_retry.sh`, which resumes from the per-epoch "last" checkpoint after a Metal graph error. M1-local had one such error at epoch 3 and was resumed from epoch 2; the cosine schedule and the best-IoU threshold were restored.

Run registry

ID	Init	Train data	Features	Budget	Selec
M0_global	authors	global benchmark	opt+DEM	authors'	—
M1_hma_official	authors (global → HMA)	HMA train	opt+DEM	authors' (80 ep, bs 8)	HMA v
M1_hma_regional_official	authors (HMA only)	HMA train	opt+DEM	authors'	HMA v

ID	Init	Train data	Features	Budget	Selec
M1_hma_local	M0 weights	165 HMA train tiles	opt+DEM	8 ep × 250 steps, bs 4 (≈1.7 % of official steps)	best v IoU (epoch
M2_cs_selftrain	M1_hma_official	15 CS pseudo-label scenes (×2) + 40 HMA replay tiles	opt+DEM	4 ep × 200 steps, bs 4, lr 2e-5	HMA v (best epoch 0.693)
M1_sarsubset_nosar	M0 weights	66 SAR-bearing HMA train tiles	opt+DEM	4 ep × 250, bs 4	HMA v SAR ti (0.689
M3_hma_sar	M0 weights + new SAR branch	same 66 tiles	opt+DEM+VV (asc, desc)	4 ep × 250, bs 4	HMA v SAR ti (0.666
M1_sarsubset_optonly	M0 weights, DEM branch dropped	same 66 tiles	optical	4 ep × 250, bs 4	HMA v SAR ti (0.551

Ablations

Question	Comparison	Evaluation data
Exp. 1: global pretrained	M0	HMA test (57 tiles); CS series
Exp. 2: HMA fine-tuning	M0 vs M1_hma_official vs M1_hma_local	HMA test; CS series
Exp. 3: CS domain adaptation	M1_hma_official vs M2	HMA test (forgetting); CS temporal consistency and inventory agreement
Exp. 4: optical only vs optical + DEM	M1_sarsubset_optonly vs M1_sarsubset_nosar (equal budgets, same tiles; weights transferred by input name)	HMA test SAR tiles (22); CS series

Question	Comparison	Evaluation data
Exp. 5: optical + DEM + SAR	M1_sarsubset_nosar vs M3_hma_sar (equal budgets, same tiles)	HMA test SAR tiles (22); CS series (approximate SAR inputs)
Sensitivity: ensemble window	core 15 Aug–15 Sep vs full 15 Aug–15 Oct	CS series (M1_hma_official)

Seeds and splits

Seed 42 (numpy, Python, TensorFlow). The official benchmark train/val/test split is unchanged.
No
HMA tile lies within 0.05° of Chhota Shigri (checked for all 277 tiles), and the nearest tiles are about 0.6° away, so M0 and M1 never saw the study site. Chhota Shigri training uses only the even-year scenes. Every analysis epoch (2017, 2019, 2021, 2023, 2025) is temporally held out.

Incidents and design revisions (chronological, all automatic re-runs)

1. **Data source.** Planetary Computer has no L1C. L1C TOA is taken from the Google public bucket
   so the inputs match the model's training domain.
2. **Cloud mask in composites.** The first composite used the raw s2cloudless probability; the scoring used the averaged mask. Both now use the averaged + dilated mask.
3. **Benchmark extraction.** Shared shard HDF5 files were corrupted when the workers were stopped.
   This was replaced by atomic per-tile files, and the test tiles were re-downloaded.
4. **Surface-condition artefact (user decision: Option A).** Single-scene areas differ by about 1 km²
   between good scenes, because snow-dusted vs exposed debris margins change the result. This led to
   the multi-date ensemble.
5. **Late-season scenes.** Scenes after 15 Sep (autumn snowfall, low sun) map 1.5–4.6 km² too little
   ice. The primary window became 15 Aug–15 Sep, with the full window kept as a sensitivity run. This
   decision was made after looking at the data and is documented as such.
6. **Robust ensemble.** Whole-scene outliers are rejected with |A − median| > max(3 × 1.4826·MAD,
   0.5 km²).

7. **Uncalibrated softmax.** GlaViTU's raw P(glacier) stays in about 0.24–0.89 (label smoothing and
   focal loss), so the 0.95/0.05 pseudo-label rule produced zero labels. An isotonic calibration was
   fitted on HMA val tiles: held-out ECE went from 0.239 to 0.011. The calibrated thresholds correspond to raw ≈ 0.70 / 0.25–0.30.
8. **Sentinel-1.** Only RTC γ^0 is available without SNAP. It was converted to approximate σ^0 with $\gamma^0 \cdot \cos \theta_inc$ (θ from the GRD annotation geolocation grid, about 34–46°). The normalised value
   distributions match the HMA benchmark within about 0.05.
9. **Minimum-observation bias (found during results assembly).** With `min_valid_obs = 2`, pixels
   cloudy in either scene of 2-scene years were set to non-glacier: −0.73 km² of glacier-like pixels in 2017, −0.29 km² in 2019, and −0.13 km² in 2021 (M2). Changed to `min_valid_obs = 1`.
   Pixels with no clear core-window observation are now gap-filled from the same year's out-of-window scenes (no cross-epoch leakage), and the rest are reported as unobserved (≤ 0.52 km², all outside the RGI outline). All ensembles, analyses and validations were re-run.
   Residual side effect: about 0.06 km² of steep 5700–5800 m slopes enters the 2017 baseline from
   snowier out-of-window scenes and later appears as "loss".
10. **Box method.** The first version had an inverted sign and a 400 m width, so it measured lateral-margin variability. It now uses half the 2002 tongue width (159 m) and 2 km length.
11. **Change uncertainty.** The calibrated threshold term is systematic (same model, every year), so
    it is propagated by recomputing ΔA at the two thresholds rather than in quadrature. The area
    trend uses weighted least squares with the independent σ.
12. **Process interruptions** at the user's request (2026-09-30, 10-02, 10-03). All stages resume from per-scene caches, per-tile files and per-epoch checkpoints, so no result depends on an
    interrupted run.